# ISTM 6227 GenAI with Business Applications
## GROUP PROJECT: GENERATIVE AI BUSINESS SOLUTION CONSULTING PROJECT

## Overview
As a major course project, your team will work on a hands-on consulting project that applies Generative AI (GenAI) tools to a real-world business context. Your team will act as a GenAI consulting team hired by a real or hypothetical organization. You will identify a meaningful business problem or opportunity and design, build, evaluate, and recommend a practical GenAI solution that creates business value.
The purpose of the project is to integrate the major concepts covered throughout the course. Rather than proposing a GenAI idea only at a conceptual level, your team should demonstrate how the solution would work through a prototype or functional demonstration and critically evaluate its value, limitations, implementation requirements, and governance risks.

## Your Task
Your team has been asked by the client organization to develop a GenAI solution for a specific business problem or opportunity. The project should follow the complete GenAI solution lifecycle:

**DEFINE → DESIGN → BUILD → DEMONSTRATE → EVALUATE → GOVERN → RECOMMEND**

## Project Components

### 1. Business Problem and Value Proposition
- Identify a real or hypothetical focal organization and a specific business problem or opportunity.
- Define the target users and relevant stakeholders.
- Explain why GenAI is appropriate for this problem rather than a conventional non-GenAI solution.
- Specify the expected business value and measurable indicators of success.

### 2. GenAI Solution and Architecture
- Describe what the proposed GenAI application will do and how users will interact with it.
- Present the overall solution architecture and workflow, including the selected model(s), data or knowledge sources, inputs, outputs, and major system components.
- Explain key design choices and how they support the business objective.

### 3. Prompt Engineering and Interaction Design
- Develop and test prompts appropriate for your application.
- Show the evolution from a baseline prompt to an improved prompt and explain why the final design performs better.
- Apply relevant techniques such as system instructions, role/persona prompting, zero-shot or few-shot prompting, structured output, and prompt enrichment.
- Include representative prompts and outputs as evidence.

### 4. Knowledge, Embeddings, and RAG
- Determine whether the application requires external or proprietary knowledge beyond the base model.
- Where appropriate, build a small knowledge base and demonstrate a Retrieval-Augmented Generation (RAG) workflow.
- Explain how documents or knowledge are represented, retrieved, and supplied to the model, and discuss how RAG may affect relevance, grounding, and hallucination.

### 5. Agentic Workflow and Automation

- Design at least one multi-step workflow or agentic feature appropriate for the business problem.
- Show how the GenAI system can perform or coordinate multiple steps, such as classification, retrieval, generation, structured extraction, tool use, decision support, or human escalation.
- Clearly identify where human oversight remains necessary.

### 6. Model Adaptation and Technical Choices

- Compare the suitability of prompting, RAG, fine-tuning, or a combination for your application.
- Justify your choice based on factors such as task requirements, data availability, accuracy, cost, privacy, scalability, and maintenance.
- Fine-tuning implementation is not required unless it is central to your project, but your team should demonstrate that you understand when and why it would be useful.

### 7. Evaluation and GenAI Operations

- Develop a systematic evaluation plan rather than relying only on a successful demonstration.
- Create a set of representative test cases or prompts and evaluate relevant dimensions such as accuracy, relevance, hallucination, consistency, response quality, latency, cost, or user experience.
- Where feasible, compare a baseline version with the final solution and summarize the results using a table or other clear evidence.
- Discuss how the system would be monitored and improved after deployment.

### 8. Governance, Risk, and Responsible AI

- Identify the major risks associated with the proposed solution, including those relevant to hallucination, bias/fairness, privacy, data security, copyright/IP, prompt injection, misuse, and human oversight.
- Propose practical safeguards and governance mechanisms.
- Explain what the organization should and should not automate with GenAI.

### 9. Implementation and Final Recommendation

- Provide a realistic implementation roadmap for the client organization.
- Discuss organizational requirements, expected benefits, major costs or constraints, and scalability.
- Conclude with a clear recommendation: Should the organization deploy the proposed solution? If so, how should it proceed?

## Possible Project Examples

Examples include, but are not limited to: a customer-service assistant, internal knowledge assistant, marketing-content system, financial-document analyzer, HR onboarding assistant, sales support tool, contract or policy review assistant, research assistant, personalized recommendation assistant, or another GenAI-enabled business application approved by the instructor.

## Project Milestones and Deliverables

| Milestone | Purpose | Expected Content |
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| 1. Project Proposal | Define the problem and initial solution | Focal organization; business problem/opportunity; target users; expected value; initial GenAI solution and architecture; proposed tools/data/knowledge sources. |
| 2. Project Update / Prototype Check | Demonstrate progress and obtain feedback | Prompt design; preliminary prototype; RAG/knowledge approach; agentic workflow; initial evaluation plan; challenges and next steps. |

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| 3. Final Project | Present an integrated, evidence-based GenAI solution | Working prototype or functional demonstration; final consulting report; evaluation evidence; governance analysis; implementation roadmap; final recommendation; presentation. |

## Final Project Requirements

- Final Consulting Report: approximately 10–15 pages, excluding appendices. The report should communicate the business problem, solution design, prototype, evaluation, governance considerations, and implementation recommendation in a professional consulting format.
- Prototype / Functional Demonstration: Your team must demonstrate a working or functional GenAI solution. The sophistication of the implementation may vary by project, but the prototype should be sufficient to test the core value proposition.
- Final Presentation: Present the project at the end of the semester. The presentation should emphasize the business problem, GenAI solution, demonstration, evaluation results, risks, and final recommendation.
- Appendix / Technical Artifacts: Include relevant prompts, sample outputs, evaluation test cases, architecture diagrams, RAG materials, system instructions, or other supporting artifacts as appropriate.
- Peer Evaluation: Because this is a group project, peer evaluation may be used to assess individual contributions.

## Suggested Final Report Structure

- Executive Summary
- 1. Business Problem and Opportunity
- 2. Proposed GenAI Solution and Business Value
- 3. GenAI Architecture and Technical Design
- 4. Prompt Engineering and Interaction Design
- 5. RAG / Knowledge and Agentic Workflow
- 6. Prototype and Demonstration
- 7. Evaluation and Results
- 8. Governance, Risk, and Responsible AI
- 9. Implementation Roadmap and Final Recommendation
- References
- Appendix: Prompts, Outputs, Test Cases, and Other Technical Artifacts

## Evaluation Criteria

| Criterion | What We Will Look For |
|---|---|
| Business problem and value | Importance and clarity of the problem; fit between GenAI and the business need; expected value. |
| Integration of course concepts | Effective application of prompt engineering, GenAI architecture, RAG/knowledge, agentic systems, model adaptation, operations, and governance as relevant. |
| Prototype and technical execution | Functionality, coherence of design, quality of demonstration, and appropriate use of GenAI tools. |
| Evaluation and critical analysis | Systematic testing, evidence-based assessment, identification of limitations, and thoughtful interpretation of results. |
| Governance and responsible AI | Identification of meaningful risks and development of practical safeguards and human oversight. |

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| Implementation and recommendation | Feasibility, scalability, organizational considerations, and strength of the final recommendation. |
| Professional communication | Organization, clarity, visual presentation, documentation, and quality of the written report and presentation. |

## Important Notes

- The project should focus on a specific business problem rather than a general discussion of Generative AI.
- A technically complex solution is not automatically a better project. Your design should be appropriate for the business problem and supported by clear reasoning and evidence.
- You are encouraged to experiment with multiple GenAI tools and models when useful, but you remain responsible for evaluating and verifying all AI-generated content.
- Your team should document the important prompts, tools, models, data/knowledge sources, and AI-generated materials used in the project.
- Additional details on due dates, presentation time, submission format, and grading weights will be announced in class and on Blackboard.

## Project Timeline

- **Project Topic / Initial Proposal: By Week 6**
  - Each team should submit a brief **one-page project idea** by Week 6.
  - This is intended as an initial proposal rather than a fully developed project plan.
  - Briefly identify the **focal organization or business context, business problem or opportunity, proposed GenAI solution, target users, and expected business value**.
  - You may also briefly indicate the GenAI tools, data, or knowledge sources you expect to use, although these choices may change as your project develops.
  - The purpose of this submission is to receive early feedback and ensure that the project is appropriately scoped for the course.
- **Final Project Presentation: December 8, 2026**
  - Each team will have **10 minutes for the presentation, followed by approximately 3 minutes of Q&A**.
  - Present the business problem, GenAI solution, prototype/demo, evaluation results, key governance considerations, and implementation recommendation.
  - Use feedback from the instructor and classmates to further improve your project.
- **Final Project Report: December 15, 2026**
  - After the presentation, each team should **incorporate relevant feedback and suggestions** into the final version of the project.
  - Submit the revised **Final Consulting Report by December 15, 2026**.
  - The final report should reflect both the work presented on December 8 and meaningful improvements made based on the feedback received.